# TELANGANA STATE-2016

# CLASS 10

# GENERAL SCIENCE, Paper-II

[Time:2 Hours 45 min.]
[Maximum Marks:40]

**Instructions:**

1. In the time duration of 2 Hours 45 minutes, 15 minutes of time isallotted to read and understand the Question paper.

2. Answer the Questions under Part-A on separate answer book.

3. Write the answers to the Questions under Part-B on the Question paper itself and attach it to the answer book of Part-A.

## Part -A

**Time: 2 Hours**
**Marks: 35**

**Instructions:**

i. Part-A comprises of three sections I, II and III.

ii. All the questions are compulsory.

iii. There is no overall choice. However, there is internal choice to theQuestions under section-III.

## SECTION –I

7x1=7

**NOTE:**

i. Answer all the following questions.

ii. Each question carries 1 mark.

iii. Write answers in 1-2 sentences for each question.

**1. What happens if the forest area decreases rapidly?**

**[Ans]-** If the forest area shrinks rapidly then there will be loss of biodiversity and Forests will not be able to maintain eco-balance.

**2. One student takes high caloric food. Another student takes less caloricfood. But both are affected with diseases. Name the diseases by whichthey are affected.**

[ans] Excess calories in your diet can lead to being overweight, obesity, diabetes, hypertension, heart disease, stroke.

low calorie foods lead to Type 2 Diabetes, osteomalacia, Kwashiorkor.

**3. Mention the two chemicals which you have used in a experiment to test the presence of starch in the leaf.**

**[Ans]-** Potassium hydroxide&Iodine solution is used to test leaves for the presence of starch.

**4. We can't expect the world without sparrows. So how should be ourconcern towards their conservation?**

1. Plan in your own balcony or garden for the availability of food grains and water for sparrows.
2. Minimize the use of mobile phones, and use anti-radiation cover to protect yourself and the environment.

3. Use of high-quality fuel in your vehicle and maintain your vehicle to minimize pollution.

**Q-5) What happens if there is no mucus in the Oesophagus?**

**[Ans]-**if there is no mucus in the Oesophagus then Food will not pass through the esophagus, there will be no process of peristalsis and then it can harden our throats.

**Q-6) Write two activities which you are performing to save the electricity.**

**[Ans]-**Two activities which we are performing to save the electricity.

1. Turn off unnecessary lights.
2. By using energy-efficient saving light bulbs & Using natural light.

**Q-7) Write the name of the nerve given in the following diagram and writeits function.**

[ans] The name of the nerve is sensory neurons

Its functions are '-

1. It carries messages towards central nervous system from nerve ending on the muscles of different sense organs.

2. It responsible for converting external stimuli from the organism's environment into internal electrical impulses.

## SECTION-II

6x2-12

**NOTE:**

i. Answer all the questions.

ii. Each question carries TWO marks.

iii. Answer the questions in 4-5 sentences.

**Q-8) Where are the valves located in human heart? Write their names.**

[ans] The heart has 4 valves: The mitral valve and tricuspid valve, which control blood flow from the atria to the ventricles. The aortic valve and pulmonary valve, which control blood flow out of the ventricles.

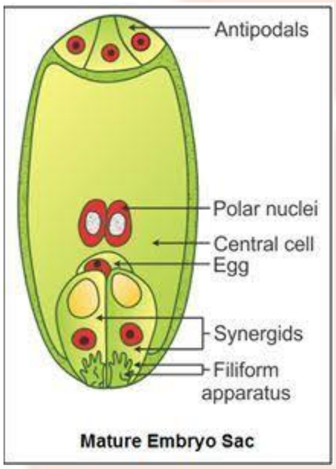
The mitral valve and tricuspid valve are located between the atria (upper heart chambers) and the ventricles (lower heart chambers). The aortic valve and pulmonic valve are located between the ventricles and the major blood vessels leaving the heart.

**Q-9) How we will get the desired useful triats with the help of two selectedtriats by using grafting method?**

[ans] The grafting method involves attachment of a graft to a rooted plant. The graft is also known as scion and the rooted plant is called as stock. The two plants are selected with respect to desirable traits. When the grafting method is performed, the

tissues of two plants are attached and allowed to grow. This leads to the production of a variety which shows the presence of both the characters. If the parent plants have traits like high yielding plant and resistant to herbicides. The new plant variety which is cultivated will have both the characters as high yield and resistant crop.

**Q-10) Draw the labelled diagram of Embryo-sac.**




**Q-11) What questions you will ask a paleontologist about fossils?**

[ans] (1) What are fossils.

(2) how dinosaur fossils prepared in the laboratory.

(3) how dinosaur fossils discovered and collected.

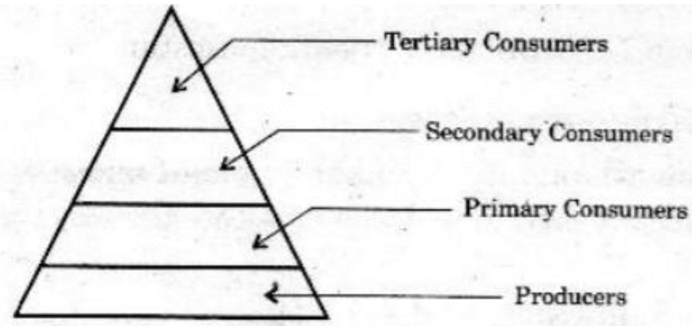
**Q-12) Write two suggestions to create awareness on groundwaterconservation.**

[ans] .  (1) dispose of chemicals properly.

(2) shut water off while brushing teeth.

(3) check for leaky faucets and have them fixed.

**Q-13) Observe the pyramid of number which is given below and answer thequestions,**




(i) As per the number of organisms in the tropic level, which group oforganisms are more in number and which are less in number?

[ans] as per the number of organisms in each of the tropic level, there is more number of producers level and less number of organisms will be tertiary consumers.

(ii) What happens if Secondary consumers disappear?

[ans] If secondary consumers disappear, then primary consumers will be increased because of no secondary consumers available which will feed primary consumers. Thus there will be decrease in tertiary consumer since there is no food available. Therefore food chain will be disturbed.

## SECTION – III

**4x4=16**

**NOTE:**

i. Answer all the questions.

ii. Each question carries4marks.

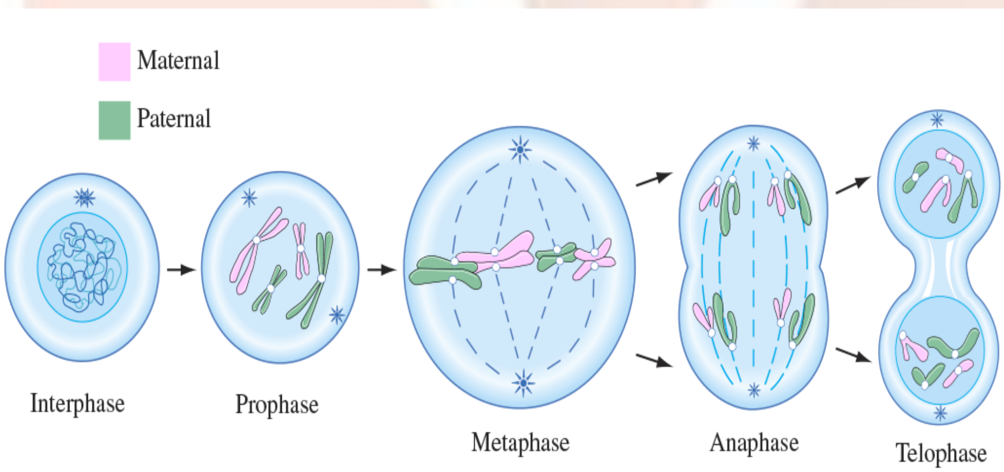
iii. There is internal choice for each question.Only one option from each question is to be attempted.

iv. Answer each question in 8-10 sentences.

**Q-14) Mention the stages of Mitosis with the help of diagrams. Explain thechanges that takes place in Prophase.**

[ans] There are five stages of mitosis that are:-

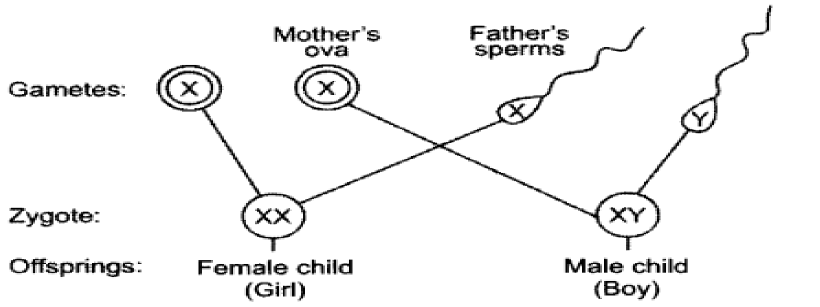
- Interphase
- Prophase
- Metaphase
- Anaphase
- Telophase




Changes that takes place in prophase:- It is the first stage of cell division in both mitosis and meiosis. Beginning after interphase. In this stage there will occur the condensation of the chromatin and the disappearance of the nucleolus.

**OR.**

Explain sex determination in humans with the help of flow-chart.




Sex determination in Human :- A sex chromosome that carries the genes for male characters is called Y chromosome and one which carries the genes for female characters is called X chromosome. We have a total of 46 chromosomes. Half of them come from the mother and the rest, from the father. Out of these 46 chromosomes, 44 are autonomies and 2 are sex chromosomes.

During gamete formation, All the eggs of a female have 22 + X chromosomes. A male produces two types of sperms—one type bears the 22 + X composition and the other, 22 + Y. Therefore, in every 100 sperms, 50 have Y chromosomes and 50 have X chromosomes. Any one of the two types of sperms can fertilize the egg. If a Y-bearing sperm fertilizes the egg, the zygote has the 44 + XY composition, and the resulting embryo grows to be a boy. When an X-bearing sperm fertilizes the egg, the resulting zygote has the 44 + XX composition. This embryo develops into a girl. All the children inherit one X chromosome from the mother.

Therefore, sex is always determined by the other sex chromosome that they inherit from the father. One who inherits the X chromosome of the father is a girl, while one who inherits the Y chromosome of the father is a boy.

**Q-15) Explain the different types of adaptations in plants with suitableexamples.**

[ans] Adaptation:- Plants adapt or adjust to their surroundings. This helps them to live and grow. A particular place or a specific habitat calls for specific conditions and adapting to such conditions helps the plants to survive. This is the reason why certain plants are found in certain areas.

Different types :-

- Plant adaptations in the desert.
- Plant adaptations in the tropical rainforest.
- Plant adaptations in the temperate forests.
- Plant adaptations in the grasslands.
- Plant adaptations in water.

Example:-

An example of adaptation behaviour of plant in water that is This is an aquatic plant. It is adapted for underwater life. This plant has its own air bubble in each leaf that provides the necessary space for the exchange of oxygen from the water to the plant. It also helps keep the seaweed upright. The leaves of underwater aquatic plants are also softer then above ground plants. This softness allows the plant to move easily with the waves without breaking.

**OR**

**What are the different stages in Urine formation? Explain whathappens in those stages.**

**[ans]**

Different stages of Urine formation and their process:- the 4 stages in urine formation are....

- Filtration: blood is filtered from the glomerulus into the bowman capsule due to great pressure. the filtrate consists of water, nutrients, electrolytes and metabolic waste products.
- Re-absorbtion: occurs in the renal tubules, reclaims useful substances for the blood
- Secretion: occurs in the renal tubules from peritubular capillaries. disposes of undesirable substances.
- Urine concentration and volume when present, ADH (antidiuretic hormone), allows water to leave via the walls of collecting ducts, altering the urines concentration and volume.

**Q-16) Write the procedure involved in the acid and leaf experiment tounderstand the concept. "how the stomach gets protected from its own acid secretions". Compare the observations with the changes that takesplace in human digestive system**.

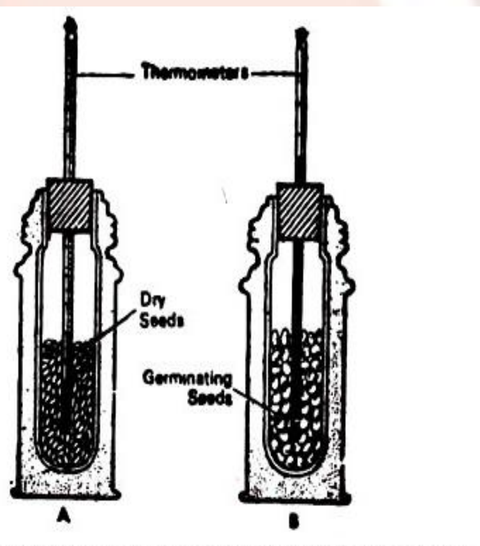
[ans] We can prove this by using a simple experiment –

Let us take 2 similar leaves and apply one leaf with petroleum jelly. Now expose both the leaves with a small amount of weak acids. After few min we will observe that the leaf with petroleum jelly color is ineffected where as the leaf without it has been effected and its color will get changed.

So with this we can conclude that mucus secreted by walls of stomach protect it from harmful effects of acids.a

**OR**

**Explain the procedure you have adopted in your school to prove thatheat is liberated during respiration. What result we will get, if youperform this experiment with dry seeds?**



*Demonstration of production of heat during respiration*

For this process we would be requiring couple of thermometer, germinating seeds, dry seeds and 2 thermal flasks.

Take both thermal flasks and add germinating seeds in flask 1 and dry seeds in flask 2. Sealed both the corks in such a way that ono heat can escape from the flask. Insert a thermometer in both the flasks.

After few min you can see that flask 2 temperature remains the same where as there is an increase in the temperature of flask 1.

Since, in flask 1 due to germination seeds the process of respiration takes place so there was an increase in temperature. Where as no such process happens in flask 2 containing dry seeds.

**Q-17) Analyse the following information and answer the questions.**

| Alkaloid | Part of the plant | Uses |
|---|---|---|
| Quinine | bark | Anti-malarial drug |
| Nicotine | leaves | Insecticide |
| Morphine | fruits | Painkiller, sedative |
| Caffeine | seeds | Central Nervous System stimulant |
| Pyrethroids | flowers | Insecticides |
| Scopolamine | fruits, flowers | Sedative |

(i) Which parts of the plants are used as alkaloids?

[ans] Bark, leaves, fruits, seeds, flowers, fruits, flowers.

(ii) What are the alkaloids which are used to control the diseases that occur in plants?

[ans] Pyrethroids

(iii) Name the parts of the plant from which we get alkaloids used as sedative.

[ans] Fruits and flowers

(iv) Name the alkaloid which is used to prevent malaria.

[ans] Quinine

**OR**

| **Vitamins** | **Resources** | **Deficiency symptoms** |
|---|---|---|
| Thiamine | Cereals, oil seeds, vegetables, milk, meat, fish, eggs | Vomitings, fits, loss of appetite, difficulty in breathing, paralysis. |
| Ascorbic acid | Green leafy vegetables, Citrus fruits, sprouts, carrot | Delay in healing of wounds, fractures of bones |
| Retinol | Leafy vegetables, carrot, tomato, pumpkin, papaya, mango, meat, fish, egg liver, milk, cod liver oil, Shark liver oil. | Night blindness, Xeropthalmia, Cornea failure, scaly skin. |
| Calciferol | Liver, egg, cod liver oil, Shark liver oil | Improper formation of bones, knock-knees, swollen wrists, delayed dentition, weak bones. |

| | | |
|---|---|---|
| Tocoferol | Fruits, vegetables, sprouts, sunflower oil | Sterility in males, abortions in females. |
| Phylloquinone | Green leafy vegetables, milk, meat, eggs | Delay in blood clotting, over-bleeding. |

(i) Which vitamins deficiency causes diseases related to bones?

**[Ans]-**Vitamin D deficiency causes diseases related to bones,

(ii) Which vitamins we get by eating fruits?

**[Ans]-**We get vitamins A, C and E by eating fruits

(iii) Which vitamin deficiency causes Paralysis? To prevent it, what type of food we have to eat?

[ans] Thiamine. To prevent from this disease we should eat Cereals, oil seeds, vegetables, milk, meat, fish, eggs

(iv) To avoid vitamin deficiency diseases, what type of food we have to eat?
[ans] Eating a healthy balance diet can prevent us from the deficiency of the disease.

## Part - B

**Marks: 10 x $^1/_2$= 5**

**Instructions:**

(i) Answer all the questions.

(ii) Each question carries $^1/_2$ mark.

(iii) Answers are to be written in question paper only.

(iv) Marks will not be awarded in any case of over-writing. rewriting or erased answers.

**I. Write the CAPITALLETTER (A,B,C,D) showing the correct answer for the following questions in the brackets provided against them.**

**Q-1) Iodine test is used to confirm the presence of ....**

(A) Fats

(B) Proteins

(C) Vitamins

(D) Carbohydrates

**[Ans] – D -** Iodine test is used to confirm the presence of starch. Starch is a type of carbohydrate,

**Q-2) In plants, the fusion of male gamate with secondary nucleus results in ....**

(A) Embryo-sac

(B) Endosperm

(C) Cotyledons

(D) Spores

**[Ans]** (B) Endosperm

In plants, the fusion of male gamate with secondary nucleus results in ....

**Q-3) Example for the reflexes in stomach is ......**

(A) Peristaltic movement.

(B) Assimilation

(C) Vomiting

(D) Digestion

[ans](A) Peristaltic movement

**Q-4) Name the body part which have division-less cells.**

(A) Brain

(B) Lungs

(C) Kidney

(D) Stomach

[ans] (A) Brain

**Q-5) The diseases caused by using mosquito repellants are ......**

(A) blood related diseases.

(B) fits.

(C) branchial diseases.

(D) Kidney related diseases.

[ans] (C) branchial diseases.

**Q-6) In plants, following is responsible for transport of water is....-**

(A) Xylem

(B) Epithelium

(C) Phloem

(D) Columnal epithelium.

[asn] (A) Xylem

**Q-7) The hormone released during hunger is .....**

(A) Adrenalin

(B) Thyroxin

(C) Leptin

(D) Ghrelin

**[Ans] – D**Ghrelin

**-**The hormone Ghrelin secreted in the stomach is responsible for hunger generating sensations.

**Q-8) The solution used to identify the presence of Oxygen in Anaerobic respiration experiment is .....**

(A) Diazene green.

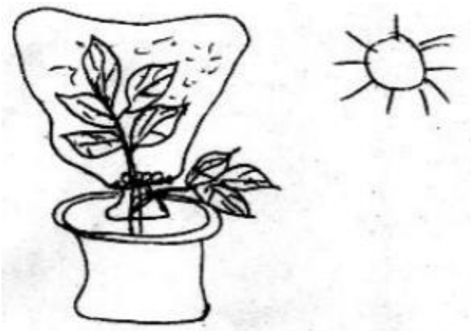
(B) Potassium hydroxide.

(C) Betadin.

(D) the rod containing Sulphur

**[Ans] – A -**The solution used to identify the presence of Oxygen in Anaerobic respiration experiment is Diazene green

**Q-9) This experiment is conducted to prove ......**



(A) Root pressure.

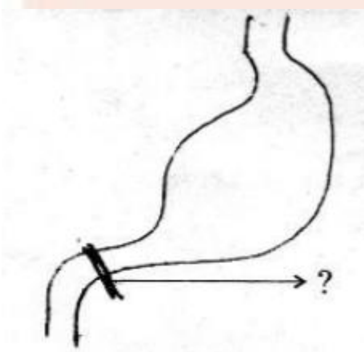
(B) Photosynthesis.

(C) Respiration.

(D) Transpiration.

**[Ans] – B -**This experiment has been done to prove photosynthesis.

**Q-10) Name the valve indicated in the picture.**




(A) Bicuspid valve.

(B) Pyloric valve.

(C) Villi.

(D) Tricuspid valve.

**[Ans] – B -**The pyloric valve, is a strong ring of smooth muscle at the end of the pyloric canal which lets food pass from the stomach to the duodenum.